8.6 Applications of Confidence Interval Estimation in Auditing

Auditing is one of the areas in business that makes widespread use of probability sampling methods in order to construct confidence interval estimates.

AUDITING

Auditing is the collection and evaluation of evidence about information related to an economic entity, such as a sole business proprietor, a partnership, a corporation, or a government agency, in order to determine and report on how well the information corresponds to established criteria.

Auditors rarely examine a complete population of information. Instead, they rely on estimation techniques based on the probability sampling methods you have studied in this text. The following list contains some of the reasons sampling is used in auditing:

- Sampling is less time-consuming.
- Sampling is less costly.
- Sampling provides an objective way of estimating the sample size in advance.
- Sampling provides results that are objective and defensible. Because the sample size is based on demonstrable statistical principles, the audit is defensible before one's superiors and in a court of law.
- Sampling provides an estimate of the sampling error and therefore allows auditors to generalize their findings to the population with a known sampling error.
- Sampling is often more accurate than other methods for drawing conclusions about large populations. Examining every item in large populations is time-consuming and therefore often subject to more nonsampling error than is statistical sampling.
- Sampling allows auditors to combine, and then evaluate collectively, samples from different individuals.

Estimating the Population Total Amount

In auditing applications, you are often more interested in developing estimates of the population **total amount** than in the population mean. Equation (8.6) shows how to estimate a population total amount.

ESTIMATING THE POPULATION TOTAL

The point estimate for the population total is equal to the population size, N, times the sample mean.

$$\text{Total} = N\bar{X} \tag{8.6}$$

Equation (8.7) defines the confidence interval estimate for the population total.

CONFIDENCE INTERVAL ESTIMATE FOR THE TOTAL

$$N\bar{X} \pm N(t_{\alpha/2})\frac{S}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} \tag{8.7}$$

where $t_{\alpha/2}$ is the critical value corresponding to an upper-tail probability of $\alpha/2$ from the t distribution with $n - 1$ degrees of freedom (i.e., a cumulative area of $1 - \alpha/2$).

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To demonstrate the application of the confidence interval estimate for the population total amount, return to the Ricknel Home Centers scenario on page 253. In addition to estimating the mean dollar amount in Section 8.2, one of the auditing tasks is to estimate the total dollar amount of all sales invoices for the month. If there are 5,000 invoices for that month and $\overline{X} = \$110.27$, then using Equation (8.6),

$$N\overline{X} = (5{,}000)(\$110.27) = \$551{,}350$$

Since $n = 100$ and $S = \$28.95$, then using Equation (8.7) with $t_{\alpha/2} = 1.9842$ for 95% confidence and 99 degrees of freedom,

$$\begin{aligned} N\overline{X} \pm N(t_{\alpha/2})\frac{S}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} &= 551{,}350 \pm (5{,}000)(1.9842)\frac{28.95}{\sqrt{100}}\sqrt{\frac{5{,}000-100}{5{,}000-1}} \\ &= 551{,}350 \pm 28{,}721.295(0.99005) \\ &= 551{,}350 \pm 28{,}435.72 \\ \$522{,}914.28 \le \text{Population total} &\le \$579{,}785.72 \end{aligned}$$

Therefore, with 95% confidence, you estimate that the total amount of sales invoices is between \$522,914.28 and \$579,785.72. Figure 8.15 shows a worksheet solution for constructing this confidence interval estimate.

Example 8.7 further illustrates the population total.

FIGURE 8.15
Excel worksheet for the confidence interval estimate of the total amount of all invoices for Ricknel Home Centers

	A	B	
1	**Total Amount of All Sales Invoices**		
2			
3	**Data**		
4	Population Size	5000	
5	Sample Mean	110.27	
6	Sample Size	100	
7	Sample Standard Deviation	28.95	
8	Confidence Level	95%	
9			
10	Intermediate Calculations		
11	Population Total	551350.00	=B4 * B5
12	FPC Factor	0.9900	=SQRT((B4-B6)/(B4 - 1))
13	Standard Error of the Total	14330.9521	=(B4 * B7 * B12)/SQRT(B6)
14	Degrees of Freedom	99	=B6 - 1
15	*t* Value	1.9842	=T.INV.2T(1 - B8, B14)
16	Interval Half Width	28435.72	=B15 * B13
17			
18	**Confidence Interval**		
19	Interval Lower Limit	522914.28	=B11 - B16
20	Interval Upper Limit	579785.72	=B11 + B16

EXAMPLE 8.7

Developing a Confidence Interval Estimate for the Population Total

An auditor is faced with a population of 1,000 vouchers and wants to estimate the total value of the population of vouchers. A sample of 50 vouchers is selected, with the following results:

$$\text{Mean voucher amount } (\overline{X}) = \$1{,}076.39$$

$$\text{Standard deviation } (S) = \$273.62$$

Construct a 95% confidence interval estimate of the total amount for the population of vouchers.

SOLUTION Using Equation (8.6), the point estimate of the population total is

$$N\overline{X} = (1{,}000)(1{,}076.39) = \$1{,}076{,}390$$

▶(*continued*)

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From Equation (8.7), a 95% confidence interval estimate of the population total amount is

$$\begin{aligned}(1{,}000)(1{,}076.39) \pm (1{,}000)(2.0096)\frac{273.62}{\sqrt{50}}\sqrt{\frac{1{,}000-50}{1{,}000-1}} \\ = 1{,}076{,}390 \pm 77{,}762.878\ (0.97517) \\ = 1{,}076{,}390 \pm 75{,}832 \\ \$1{,}000{,}558 \le \text{Population total} \le \$1{,}152{,}222\end{aligned}$$

Therefore, with 95% confidence, you estimate that the total amount of the vouchers is between $1,000,558 and $1,152,222.

Difference Estimation

Auditors use **difference estimation** when they believe that errors exist in a set of items and he or she wants to estimate the magnitude of the errors based only on a sample. To use difference estimation, follow these steps:

1. Determine the sample size required.
2. Calculate the differences between the values reached during the audit and the original values recorded. The difference in value i, denoted D_i, is equal to 0 if the auditor finds that the original value is correct, is a positive value when the audited value is larger than the original value, and is negative when the audited value is smaller than the original value.
3. Calculate the mean difference in the sample, $\overline{D}$, that Equation (8.8) defines, by dividing the total difference by the sample size.

MEAN DIFFERENCE

$$\overline{D} = \frac{\sum_{i=1}^{n} D_i}{n} \tag{8.8}$$

where D_i = audited value − original value

4. Calculate the standard deviation of the differences, S_D, that Equation (8.9) defines. *Remember that any item that is not in error has a difference value of 0.*

STANDARD DEVIATION OF THE DIFFERENCE

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}} \tag{8.9}$$

5. Use Equation (8.10) to construct a confidence interval estimate of the total difference in the population.

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CONFIDENCE INTERVAL ESTIMATE FOR THE TOTAL DIFFERENCE

$$N\overline{D} \pm N(t_{\alpha/2})\frac{S_D}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} \quad \textbf{(8.10)}$$

where $t_{\alpha/2}$ is the critical value corresponding to an upper-tail probability of $\alpha/2$ from the t distribution with $n - 1$ degrees of freedom (i.e., a cumulative area of $1 - \alpha/2$).

The auditing procedures for Ricknel Home Centers require a 95% confidence interval estimate of the difference between the audited dollar amounts on the sales invoices and the amounts originally entered into the integrated inventory and sales information system. The data are collected by taking a sample of 100 sales invoices. The results of the sample are organized and stored in the **Plumbing Invoices** workbook. There are 12 invoices in which the audited dollar amount on the sales invoice and the amount originally entered into the integrated inventory management and sales information system are different. These 12 differences are

\$9.03 \$7.47 \$17.32 \$8.30 \$5.21 \$10.80 \$6.22 \$5.63 \$4.97 \$7.43 \$2.99 \$4.63

The other 88 invoices are not in error. Each of their *differences* is 0. Thus, to analyze the data, one calculates

In the numerator, there are 100 differences. Each of the last 88 is equal to $(0 - 0.9)^2$.

$$\overline{D} = \frac{\sum_{i=1}^{n} D_i}{n} = \frac{90}{100} = 0.90$$

and

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}}$$

$$= \sqrt{\frac{(9.03 - 0.9)^2 + (7.47 - 0.9)^2 + \cdots + (0 - 0.9)^2}{100 - 1}}$$

$$S_D = 2.752$$

Using Equation (8.10), you construct the 95% confidence interval estimate for the total difference in the population of 5,000 sales invoices, as follows:

$$(5{,}000)(0.90) \pm (5{,}000)(1.9842)\frac{2.752}{\sqrt{100}}\sqrt{\frac{5{,}000 - 100}{5{,}000 - 1}}$$

$$= 4{,}500 \pm 2{,}702.91$$

$$\$1{,}797.09 \leq \text{Total difference} \leq \$7{,}202.91$$

Thus, the auditor estimates with 95% confidence that the total difference between the sales invoices, as determined during the audit, and the amount originally entered into the accounting system is between \$1,797.09 and \$7,202.91. Figure 8.16 shows the worksheet results for these data.

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FIGURE 8.16
Excel worksheet for the total difference between the invoice amounts found during audit and the amounts entered into the accounting system for Ricknel Home Centers

	A	B	
1	**Total Difference In Actual and Entered**		
2			
3	**Data**		
4	Population Size	5000	
5	Sample Size	100	
6	Confidence Level	95%	
7			
8	Intermediate Calculations		
9	Sum of Differences	90	=SUM(DIFFERENCES!A:A)
10	Average Difference in Sample	0.9	=B9/B5
11	Total Difference	4500	=B4 * B10
12	Standard Deviation of Differences	2.7518	=SQRT(E15)
13	FPC Factor	0.9900	=SQRT((B4-B5)/(B4-1))
14	Standard Error of the Total Diff.	1362.2064	=(B4 * B12 * B13)/SQRT(B5)
15	Degrees of Freedom	99	=B5 - 1
16	*t* Value	1.9842	=T.INV.2T(1 - B6, B15)
17	Interval Half Width	2702.9129	=B16*B14
18			
19	**Confidence Interval**		
20	Interval Lower Limit	1797.09	=B11 - B17
21	Interval Upper Limit	7202.91	=B11 + B17

In the previous example, all 12 differences are positive because the audited amount on the sales invoice is more than the amount originally entered into the accounting system. In some circumstances, one could have negative errors. Example 8.8 illustrates such a situation.

EXAMPLE 8.8

Difference Estimation

Returning to Example 8.7, suppose that 14 vouchers in the sample of 50 vouchers contain errors. The values of the 14 errors are listed below and stored in Difference Test. Two of the differences are negative.

$75.41	$38.97	$108.54	−$37.18	$62.75	$118.32	−$88.84
$127.74	$55.42	$39.03	$29.41	$47.99	$28.73	$84.05

Construct a 95% confidence interval estimate for the total difference in the population of 1,000 vouchers.

SOLUTION For these data,

$$\overline{D} = \frac{\sum_{i=1}^{n} D_i}{n} = \frac{690.34}{50} = 13.8068$$

and

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}}$$

$$= \sqrt{\frac{(75.41 - 13.8068)^2 + (38.97 - 13.8068)^2 + \cdots + (0 - 13.8068)^2}{50 - 1}}$$

$$= 37.427$$

▶(*continued*)

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Using Equation (8.10), construct the confidence interval estimate for the total difference in the population, as follows:

$$(1{,}000)(13.8068) \pm (1{,}000)(2.0096)\frac{37.427}{\sqrt{50}}\sqrt{\frac{1{,}000 - 50}{1{,}000 - 1}}$$

$$= 13{,}806.8 \pm 10{,}372.4$$

$$\$3{,}434.40 \leq \text{Total difference} \leq \$24{,}179.20$$

Therefore, with 95% confidence, you estimate that the total difference in the population of vouchers is between $3,434.40 and $24,179.20.

One-Sided Confidence Interval Estimation of the Rate of Noncompliance with Internal Controls

Organizations use internal control mechanisms to ensure that individuals act in accordance with company guidelines. For example, Ricknel Home Centers requires that an authorized warehouse-removal slip be completed before goods are removed from the warehouse. During the monthly audit of the company, the auditing team is charged with the task of estimating the proportion of times goods were removed without proper authorization. This is referred to as the *rate of noncompliance with the internal control.*

To estimate the rate of noncompliance, auditors take a random sample of sales invoices and determine how often merchandise was shipped without an authorized warehouse-removal slip. The auditors then compare their results with a previously established tolerable exception rate, which is the maximum allowable proportion of items in the population not in compliance. When estimating the rate of noncompliance, it is commonplace to use a **one-sided confidence interval**. That is, the auditors estimate an upper bound on the rate of noncompliance. Equation (8.11) defines a one-sided confidence interval for a proportion.

ONE-SIDED CONFIDENCE INTERVAL FOR A PROPORTION

$$\text{Upper bound} = p + Z_\alpha\sqrt{\frac{p(1-p)}{n}}\sqrt{\frac{N-n}{N-1}} \qquad \textbf{(8.11)}$$

where Z_α = the value corresponding to a cumulative area of $(1 - \alpha)$ from the standardized normal distribution (i.e., a right-tail probability of α).

If the tolerable exception rate is higher than the upper bound, the auditor concludes that the company is in compliance with the internal control. If the upper bound is higher than the tolerable exception rate, the auditor has failed to prove that the company is in compliance. The auditor may then request a larger sample.

Consider the example of 400 sales invoices that were selected from a population of 10,000 invoices, in which 20 were found to be in violation of an internal control. If the tolerable exception rate for this internal control is 6%, what should one conclude?

Using a 95% level of confidence, $p = 20/400 = 0.05$ and $Z_\alpha = 1.645$, then by using Equation (8.11),

$$\text{Upper bound} = p + Z_\alpha\sqrt{\frac{p(1-p)}{n}}\sqrt{\frac{N-n}{N-1}} = 0.05 + 1.645\sqrt{\frac{0.05(1-0.05)}{400}}\sqrt{\frac{10{,}000-400}{10{,}000-1}}$$

$$= 0.05 + 1.645(0.0109)(0.98) = 0.05 + 0.0176 = 0.0676$$

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Therefore, with 95% confidence, the rate of noncompliance is less than 6.76%. Because the tolerable exception rate is 6%, the rate of noncompliance may be too high for this internal control. The noncompliance rate for the population may be higher than the rate deemed tolerable. Therefore, one should request a larger sample.

In many cases, the auditor is able to conclude that the rate of noncompliance with the company's internal controls is acceptable. Example 8.9 illustrates such an occurrence.

EXAMPLE 8.9

Estimating the Rate of Noncompliance

A large electronics firm writes 1 million checks a year. An internal control policy for the company is that the authorization to sign each check is granted only after an invoice has been initialed by an accounts payable supervisor. The company's tolerable exception rate for this control is 4%. If control deviations are found in 8 of the 400 invoices sampled, what should the auditor do? To solve this, use a 95% level of confidence.

SOLUTION The auditor constructs a 95% one-sided confidence interval for the proportion of invoices in noncompliance and compares this to the tolerable exception rate. Using Equation (8.11), $p = 8/400 = 0.02$, and $Z_\alpha = 1.645$ for 95% confidence,

$$\text{Upper bound} = p + Z_\alpha\sqrt{\frac{p(1-p)}{n}}\sqrt{\frac{N-n}{N-1}} = 0.02 + 1.645\sqrt{\frac{0.02(1-0.02)}{400}}\sqrt{\frac{1{,}000{,}000-400}{1{,}000{,}000-1}}$$

$$= 0.02 + 1.645(0.007)(0.9998) = 0.02 + 0.0115 = 0.0315$$

The auditor concludes with 95% confidence that the rate of noncompliance is less than 3.15%. Because this is less than the tolerable exception rate, the auditor concludes that the internal control compliance is adequate. In other words, the auditor is more than 95% confident that the rate of noncompliance is less than 4%.

PROBLEMS FOR SECTION 8.6

LEARNING THE BASICS

8.70 A sample of 25 is selected from a population of 500 items. The sample mean is 25.7, and the sample standard deviation is 7.8. Construct a 99% confidence interval estimate for the population total.

8.71 Suppose that a sample of 200 (stored in Item Sample) is selected from a population of 10,000 items. Of these, 10 items are found to have the following errors:

13.76	42.87	34.65	11.09	14.54
22.87	25.52	9.81	10.03	15.49

Construct a 95% confidence interval estimate for the total difference in the population.

8.72 If $p = 0.04$, $n = 300$, and $N = 5{,}000$, calculate the upper bound for a one-sided confidence interval estimate for the population proportion, π, using the following levels of confidence:

a. 90%
b. 95%
c. 99%

APPLYING THE CONCEPTS

8.73 A stationery store wants to estimate the total retail value of the 1,000 greeting cards it has in its inventory. Construct a 95% confidence interval estimate for the population total value of all greeting cards that are in inventory if a random sample of 100 greeting cards indicates a mean value of $2.55 and a standard deviation of $0.44.

8.74 The personnel department of a large corporation employing 3,000 workers wants to estimate the family dental expenses of its employees to determine the feasibility of providing a dental insurance plan. A random sample of 10 employees (stored in the file Dental) reveals the following family dental expenses (in dollars) for the preceding year:

110 362 246 85 510 208 173 425 316 179

Construct a 90% confidence interval estimate for the total family dental expenses for all employees in the preceding year.

8.75 A branch of a chain of large electronics stores is conducting an end-of-month inventory of the merchandise in stock. There were 1,546 items in inventory at that time. A sample of 50 items was randomly selected and an audit was conducted with the following results:

Value of Merchandise

$\bar{X} = \$252.28 \quad S = \$93.67$

Construct a 95% confidence interval estimate for the total value of the merchandise in inventory at the end of the month.

8.76 A customer in the wholesale garment trade is often entitled to a discount for a cash payment for goods. The amount of discount varies by vendor. A sample of 150 items selected from a population of 4,000 invoices at the end of a period of time (stored in Discount) revealed that in 13 cases, the customer failed to take the discount

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to which he or she was entitled. The amounts (in dollars) of the 13 discounts that were not taken were as follows:

6.45	15.32	97.36	230.63	104.18	84.92	132.76
66.12	26.55	129.43	88.32	47.81	89.01	

Construct a 99% confidence interval estimate for the population total amount of discounts not taken.

8.77 Econe Dresses is a small company that manufactures women's dresses for sale to specialty stores. It has 1,200 inventory items, and the historical cost is recorded on a first-in, first-out (FIFO) basis. In the past, approximately 15% of the inventory items were incorrectly priced. However, any misstatements were usually not significant. A sample of 120 items was selected, and the historical cost of each item was compared with the audited value. The results indicated that 15 items differed in their historical costs and audited values. The data for the 15 items appear in the following table and are stored in FIFO.

Item Number	Historical Cost ($)	Audited Value ($)	Item Number	Historical Cost ($)	Audited Value ($)
5	261	240	60	21	210
9	87	105	73	140	152
17	201	276	86	129	112
18	121	110	95	340	216
28	315	298	96	341	402
35	411	356	107	135	97
43	249	211	119	228	220
51	216	305			

Construct a 95% confidence interval estimate for the total population difference in the historical cost and audited value.

8.78 Tom and Brent's Alpine Outfitters conducts an annual audit of its financial records. An internal control policy for the company is that a check can be issued only after the accounts payable manager initials the invoice. The tolerable exception rate for this internal control is 0.04. During an audit, a sample of 300 invoices is examined from a population of 10,000 invoices, and 11 invoices are found to violate the internal control.

a. Calculate the upper bound for a 95% one-sided confidence interval estimate for the rate of noncompliance.

b. Based on (a), what should the auditor conclude?

8.79 An internal control policy for Rhonda's Online Fashion Accessories requires a quality assurance check before a shipment is made. The tolerable exception rate for this internal control is 0.05. During an audit, 500 shipping records were sampled from a population of 5,000 shipping records, and 12 were found that violated the internal control.

a. Calculate the upper bound for a 95% one-sided confidence interval estimate for the rate of noncompliance.

b. Based on (a), what should the auditor conclude?

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EG8.6 APPLICATIONS of CONFIDENCE INTERVAL ESTIMATION in AUDITING

Estimating the Population Total Amount

Example Compute the Figure 18.15 confidence interval estimate of the total of all sales invoices.

PHStat Use **Estimate for the Population Total**

For the example, select **PHStat → Confidence Intervals → Estimate for the Population Total**. In the procedure's dialog box:

1. Enter **5000** as the **Population Size**.
2. Enter **95** as the **Confidence Level** percentage.
3. Click **Sample Statistics Known** and enter **100** as the **Sample Size, 110.27** as the **Sample Mean**, and **28.95** as the **Sample Standard Deviation**.
4. Enter a **Title** and click **OK**.

Workbook Use the **COMPUTE worksheet** of the **CIE Total workbook**, shown in Figure 8.15, as a template for computing confidence interval estimates for the population total. The worksheet contains the data for the example. In cell B15, the worksheet uses **T.INV.2T**(1–***confidence level, degrees of freedom***) to determine the *t* value. To compute confidence interval estimates for other problems, change the **Population Size**, **Sample Mean**, **Sample Size**, **Sample Standard Deviation**, and **Confidence Level** values in cells B4 through B8.

If you use an Excel version older than Excel 2010, use the COMPUTE_OLDER worksheet that uses the equivalent older function **TINV**.

Difference Estimation

Example Compute the Figure 18.16 confidence interval estimate of the total difference for the Ricknel Home Centers auditing example.

PHStat Use **Estimate for the Total Difference**

For the example, open to the **DATA worksheet** of the **PlumbInv workbook**. Select **PHStat → Confidence Intervals → Estimate for the Total Difference**. In the procedure's dialog box:

1. Enter **100** as the **Sample Size**.
2. Enter **5000** as the **Population Size**.
3. Enter **95** as the **Confidence Level** percentage.
4. Enter **A1:A13** as the **Differences Cell Range** and check **First cell contains label**.
5. Enter a **Title** and click **OK**.

This procedure creates a worksheet containing the results and a second worksheet that contains computations that help to calculate the standard deviation of the differences. (The *Workbook* section that follows discusses contents of the second worksheet.)

Workbook Use the **COMPUTE worksheet** of the **CIE Total Difference workbook**, shown in Figure 8.16, as a template for computing confidence interval estimates for the total difference. The worksheet contains the data for the example, including the difference data from PlumbInv.

In the worksheet, the **Sum of Differences** in cell B9 and the **Standard Deviation of Differences** in cell B12 rely on computations found in the **DIFFERENCES worksheet**. To examine the formulas used in that worksheet, open to the **DIFFERENCES_FORMULAS worksheet** in the same workbook.

Computing the total difference confidence interval estimate for other problems requires changes to both the COMPUTE and DIFFERENCES worksheets. First, in the COMPUTE worksheet, change the population size, sample size, and confidence level in cells B4 through B6, respectively. Then, in the DIFFERENCES worksheet, enter the differences in column A and adjust the column B formulas so that there is a column B formula for each difference listed. If there are more than 12 differences, select cell B13 and copy down through all the rows. If there are fewer than 12 differences, delete formulas in column B from the bottom up, starting with cell B13, until there are as many formulas as there are difference values.

If you use an Excel version older than Excel 2010, use the COMPUTE_OLDER worksheet that uses the equivalent older function **TINV**.

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